

Daniel Andrade Schuch

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Research Profile

Atmospheric scientist from Brazil with expertise in weather and air quality modeling, data analysis, and computational tool development. Experienced in the application, evaluation, and development of numerical models, integrating simulations with observational datasets (including satellite remote sensing) to investigate atmospheric composition, air quality, and climate processes. Proven track record of leading research projects and developing reproducible, high-performance computational workflows. Active contributor to open-source initiatives, with the ability to translate complex scientific challenges into practical and robust solutions. Research interests include atmospheric dynamics, boundary-layer processes, turbulence, transport phenomena, aerosols, and atmospheric chemistry. Committed to applying advanced modeling and data analysis techniques to support decision-making regarding atmospheric environmental challenges related to climate and air quality.

Online: [resume](#) - [scholar.google](#) - [orcid](#) - [researchgate](#) - [Web of Science](#) - [github](#) - [Linkedin](#)

Education

DSC in Sciences and Spatial Technology 2013-2017

Thesis: *Atmospheric model for forecasting and transporting effluent gases from rocket launches applied to the Alcântara Launch Center.*

Technological Institute of Aeronautics (ITA), São Paulo, Brazil

MSc in Meteorology, Micrometeorology 2009-2011

Thesis: *Simulation of pollutant dispersion in the atmosphere using the integral transform technique for an arbitrary source.*

University of Pelotas (UFPEL), Pelotas, Brazil

BSc in Meteorology 2005-2009

University of Pelotas (UFPEL), Pelotas, Brazil

Experience

Postdoctoral Research Associate 2020-2026

Northeastern University, Boston-MA, U.S.

Mentoring of PhD candidates. Application, evaluation and development of Air Quality models for multiple projects and regions at CASCADE (Clean Air Smart City and Digital Earth) lab, across regions including Brazil, US, Africa, and Dubai.

- Brazil: Multi-year WRF-Chem projections of air quality
- US: Built GHG emission frameworks for WRF-GHG, and application and evaluation of UFS-CMAQ
- Africa: Application and optimization of WRF-Chem to simulate PM_{2.5}
- Dubai: Applied CAMx with advanced tools for source apportionment and emissions scenario simulation

Instructor for WRF-Chem model

2024-2024

World Meteorological Organization (WMO)

WRF-Chem model training school on air quality prediction and forecasting for “Training Course on Seamless Prediction of Air Pollution in Africa”

- Designed model domains, prepared inputs, and performed sensitivity analysis for WRF-Chem during a training course on air quality forecasting in Africa
- Delivered online instruction and led hands-on simulation sessions for international trainees

Postdoctoral Research

2018-2019

University of São Paulo, São Paulo-SP, Brazil

Study of the impact on air quality of emission scenarios in the formation of the secondary pollutants: ozone and particulate matter in Brazilian metropolitan regions using numerical models.

- Investigated secondary pollutant (O_3 and $PM_{2.5}$) formation under varied emission and climate scenarios using WRF-Chem
- Generated insights into future air quality trends across Brazilian metropolitan regions

Postdoctoral Research

2017-2018

University of São Paulo, São Paulo-SP, Brazil

Research focused on obtaining a complete diagnosis of the impact of vehicle fuels on air quality in urban regions in the southeast of Brazil and adjacent areas, through atmospheric modelling and scenario studies.

- Analyzed the impact of vehicle fuels on air quality in São Paulo through scenario modeling and WRF-Chem simulations using street-level emissions
- Environmental Responsibility Award, XIV Brazilian Association of Automotive Engineering, 2021

Specialist external consultant

2016-2016

ENEVA LTD., Rio de Janeiro-RJ, Brazil

Perform simulations of the dispersion of sulfur dioxide (SO_2) from the thermoelectric power generation plant at Porto do Itaqui located in São Luís (Maranhão) using WRF-Chem and analysis of the results. The analysis was delivered in the form of media (images and animation) and a technical report together with the technical opinion.

- Simulated dispersion of SO_2 from a thermoelectric plant using WRF-Chem; produced technical reports, visualizations, and animations for stakeholders

Researcher DTI-2

2011-2013

INPE, São José dos Campos-SP, Brazil

The research focused on determining turbulent flow behavior in clearing regions and the mechanisms of atmospheric interaction between clearings and the Amazon forest, such as recirculation, boundary layer structure and formation of organized structures for this type of flow regime.

- Modeled turbulent flow and boundary layer dynamics for clearings in the Amazon forest using LES and wind tunnel studies

Peer-reviewed journal articles

25. **Daniel Schuch**, and Y. Zhang; Application of WRF-CAMx over West Asia, Part II: Ozone Formation Regimes and Process Analysis. *Climate*, 2026. <https://doi.org/10.3390/cli14060116>
24. **Daniel Schuch**, Y. Zhang, and K.Farzad; Application of WRF-CAMx over West Asia, Part I: Meteorological and Air Quality Model Evaluation. *Climate*, 2026. <https://doi.org/10.3390/cli14060128>
23. **Daniel Schuch**, Y. Zhang, S. Ibarra-Espinosa,c, M. F. Andradede, M. Gavidia-Calderónd, and M. L. Belle; Multi-Year Evaluation and Application of the WRF-Chem Model for Two Major Urban Areas in Brazil - Part II: Multi-Year Evaluation and Urban-centric Analysis. *Atmospheric Environment*, 2026. <https://doi.org/10.1016/j.atmosenv.2025.121632>
22. **Daniel Schuch**, Y. Zhang, S. Ibarra-Espinosa,c, M. F. Andradede, M. Gavidia-Calderónd, and M. L. Belle; Multi-Year Application and Evaluation of the WRF-Chem Model for Two Major Urban Areas in Brazil - Part I: Initial Application and Model Improvement. *Atmospheric Environment*, 2026. <https://doi.org/10.1016/j.atmosenv.2025.121577>
21. **Schuch, Daniel**. eva3dm: A R-package for model evaluation of 3D weather and air quality models. *Journal of Open Source Software*, 10(108), 7797, 2025. <https://doi.org/10.21105/joss.07797>
20. Seulkee Heo, **Daniel Schuch**; WASHINGTON LEITE JUNGER; YANG ZHANG; MARIA DE FATIMA ANDRADE; MICHELLE L. BELL. The impact of exposure assessment on associations between air pollution and cardiovascular mortality risks in the city of Rio de Janeiro, Brazil. Submitted to Environmental Research, 2024. <http://doi.org/10.1016/j.envres.2024.120150>
19. PACHON, J. E., Opazo, M., Lichtig, P., Hunneus, N., Bouarar, I., Brasseur, G., Li, C. W. Y., Flemming, J., Menut, L., Menares, C., Gallardo, L., Gauss, M., Sofiev, M., Kouznetsov, R., Palamarchuk, J., Dawidowski, L., Rojas, N. Y., Andrade, M. D. F., Gavidia-Calderón, M. E., Delgado Peralta, A. H., and **Schuch, D.**: Air quality modeling intercomparison and multi-scale ensemble chain for Latin America, EGUsphere, 2024. <https://doi.org/10.5194/egusphere-2024-815>
18. GAVIDIA-CALDERÓN, MARIO; **SCHUCH, DANIEL**; VARA-VELA, ANGEL; INOUE, RITA; FREITAS, EDMILSON D.; ALBUQUERQUE, TACIANA TOLEDO DE A.; ZHANG, YANG; ANDRADE, MARIA DE FATIMA; BELL, MICHELLE L. Air quality modeling in the metropolitan area of São Paulo, Brazil: A review. *ATMOSPHERIC ENVIRONMENT*, v.319, p.120301, 2024. <http://doi.org/10.1016/j.atmosenv.2023.120301>
17. LYU, XIAOPU; LI, KE; GUO, HAI; MORAWSKA, LIDIA; ZHOU, BEINING; ZEREN, YANGZONG; JIANG, FEI; CHEN, CHANGHONG; GOLDSTEIN, ALLEN H.; XU, XIAOBIN; WANG, TAO; LU, XIAO; ZHU, TONG; QUEROL, XAVIER; CHATANI, SATORU; LATIF, MOHD TALIB; **SCHUCH, DANIEL**; SINHA, VINAYAK; KUMAR, PRASHANT; MULLINS, BENJAMIN; SEGUEL, RODRIGO; SHAO, MIN; XUE, LIKUN; WANG, NAN; CHEN, JIANMIN; GAO, JIAN; CHAI, FAHE; SIMPSON, ISOBEL; SINHA, BAERBEL; BLAKE, DONALD R. A synergistic ozone-climate control to address emerging ozone pollution challenges. *One Earth*, v.6, p.964 - 977, 2023. <http://doi.org/10.1016/j.oneear.2023.07.004>
16. PENDHARKAR, JAYANT; FIGUEROA, SILVIO NILO; VARA-VELA, ANGEL; KRISHNA, R. PHANI MURALI; **SCHUCH, DANIEL**; KUBOTA, PAULO YOSHIO; ALVIM, DÉBORA SOUZA; VENDRASCO, EDER PAULO; GOMES, HELBER BARROS; NOBRE, PAULO; HERDIES, DIRCEU LUÍS. Towards Unified Online-Coupled Aerosol Parameterization for the Brazilian Global Atmospheric Model (BAM): Aerosol-Cloud Microphysical-Radiation Interactions. *Remote Sensing*, v.15, p.278, 2023. <http://doi.org/doi:10.3390/rs15010278>
15. MEDEIROS, LUIZ E.; FISCH, G.; ACEVEDO, OTÁVIO C.; COSTA, FELIPE D.; IRIART, PAULO G.; ANABOR, VAGNER; **SCHUCH, Daniel**. Low-Level Atmospheric Flow at the Central North Coast of Brazil. *BOUNDARY-LAYER METEOROLOGY*, v.179, p.1 - 15, 2021. <http://doi.org/10.1007/s10546-021-00625-1>
14. **SCHUCH, DANIEL**; ANDRADE, MARIA DE FATIMA; ZHANG, YANG; DIAS DE FREITAS, EDMILSON; BELL, MICHELLE L. Short-Term Responses of Air Quality to Changes in Emissions under the Representative Concentration Pathway 4.5 Scenario over Brazil. *Atmosphere*, v.11, p.799, 2020. <https://doi.org/10.3390/atmos11080799>
13. **DANIEL SCHUCH**; Sergio Ibarra Espinosa; Leila Droprinchinski Martins; Vanessa Silveira Barreto Carvalho; Bruna Ferreira Ramin; Jayne Souza Silva; Jorge Alberto Martins; Edmilson Dias de Freitas and Maria de Fatima Andrade. A two decades study on ozone variability and trend over the main urban areas of the São Paulo state – Brazil. *Environmental Science and Pollution Research* (2019). <https://doi.org/10.1007/s11356-019-06200-z>
12. **SCHUCH, DANIEL**, DIAS DE FREITAS, EDMILSON, IBARRA-ESPINOSA ANDRADE, M. F. EmissV: A preprocessor for WRF-Chem model. *Journal of Atmospheric Science Research*. v.1, p.1 - 5, 2019. <https://doi.org/10.30564/jasr.v1i1.347>
11. IBARRA-ESPINOSA, SERGIO; **SCHUCH, DANIEL**; DIAS DE FREITAS, EDMILSON eixport: An R package to export emissions to atmospheric models. *Journal of Open-Source Software*, v.3, p.607 - 1, 2018.

<https://doi.org/10.21105/joss.00607>

10. **SCHUCH, DANIEL**; IBARRA-ESPINOSA, SERGIO; DIAS DE FREITAS, EDMILSON EmissV: An R package to create vehicular and other emissions for air quality models. *Journal of Open Source Software*, v.3, p.662 - 6, 2018. <https://doi.org/10.21105/joss.00662>

9. LEITE, HENRIQUE FANINI; AVELAR, ANA CRISTINA; DE ABREU, LEANDRA; **SCHUCH, DANIEL**; CAVALIERI, ANDRÉ Proper Orthogonal Decomposition and Spectral Analysis of a Wall-Mounted Square Cylinder Wake. *JOURNAL OF AEROSPACE TECHNOLOGY AND MANAGEMENT (ONLINE)*, v.10, p.1 - 1, 2018. <https://doi.org/10.5028/jatm.v10.867>

8. **SCHUCH, Daniel**; FISCH, G. Rocket emissions representation in atmospheric air quality models: The short-range atmospheric transport and reaction of gases released by solid propellant engines. *Meteorological Applications*, v.25, p.1 - 10, 2018. <https://doi.org/10.1002/met.1750>

7. CORRÊA, CLEBER SOUZA; **SCHUCH, DANIEL**; DE QUEIROZ, ANTONIO PAULO; FISCH, GILBERTO FERNANDO; CORRÊA, FELIPE DO NASCIMENTO; COUTINHO, MARIANE MENDES The Long-Range Memory and the Fractal Dimension: A Case Study for Alcântara. *JOURNAL OF AEROSPACE TECHNOLOGY AND MANAGEMENT (ONLINE)*, v.9, p.461 - 468, 2017. <https://doi.org/10.5028/jatm.v9i4.683>

6. **SCHUCH, DANIEL**; FISCH, GILBERTO. The Use of an Atmospheric Model to Simulate the Rocket Exhaust Effluents Transport and Dispersion for the Centro de Lançamento de Alcântara. *JOURNAL OF AEROSPACE TECHNOLOGY AND MANAGEMENT (ONLINE)*, v.9, p.137 - 146, 2017. <https://doi.org/10.5028/jatm.v9i2.740>

5. MEDEIROS, LUIZ E.; FISCH, G.; IRIART, P.; COSTA, F. D.; OLIVEIRA, D. W.; **SCHUCH, Daniel**. Padrões de vento a nível de superfície para região da costa norte do Brasil (Surface-level wind patterns for the northern Brazilian coast region). *CIÊNCIA E NATURA*. v.38, p.383 - 387, 2016.

4. **SCHUCH, DANIEL**; MARCIOTTO, EDSON R.; ACEVEDO, OTÁVIO; FISCH, GILBERTO; AVELAR, ANA CRISTINA. Estudo do escoamento turbulento atmosférico utilizando ensaios em túnel de vento e simulação numérica (Study of atmospheric turbulent flow using wind tunnel and numerical simulation). *Revista Brasileira de Meteorologia*, v.29, p.338 - 350, 2014. <https://doi.org/10.1590/0102-778620130580>

3. DEMARCO, G.; PUHALES, F. S.; COSTA, F. D.; **SCHUCH, Daniel**; FARIA, A. F.; AVELAR, A. C.; FISCH, G.; ACEVEDO, O. Análise da influência da estratificação sobre perfis de vento e temperatura em experimentos de túnel de vento (Analysis of the influence of stratification on wind profiles and temperature in wind tunnel experiments). *Ciência e Natura*, v.Esp, p.172 - 174, 2013.

2. **SCHUCH, Daniel**; MARCIOTTO, E. R.; FISCH, G.; AVELAR, A. C. Perfis de vento e turbulência atmosférica simulados em túnel de vento (Wind profiles and atmospheric turbulence simulated in wind tunnel). *Ciência e Natura*, v.Esp, p.159 - 161, 2013. <https://doi.org/10.5902/2179460X11588>

1. **SCHUCH, Daniel**; WEYMAR, Guilherme J.; FURTADO, Igor C.; QUADROS, R. S.; BUSKE, D. Simulação da dispersão de poluentes na camada limite atmosférica incluindo o efeito da deposição seca no solo (Simulation of dispersion of pollutants in the atmospheric boundary layer including the effect of dry deposition on soil). *Ciência e Natura*. v.33, p.127 - 130, 2011.

OSS Contribution

I'm a creator and contributor of Open Source Software R-packages in CRAN (Comprehensive R Archive Network, official R repository) and github.

- [EmissV](#) - Tools to calculate and process emissions for air quality models;
- [eva3dm](#) - Evaluation of 3D Meteorological and air quality models;
- [eixport](#) - Tools to export emissions to air quality atmospheric models;
- [vein](#) - Model that estimates vehicular emissions by top-down and bottom-up methods;
- [qualR](#) - Tools to download air pollutants and meteorological information.

Skills

Modeling Skills

- **Air Quality Models** (Expert): WRF-Chem (emissions, boundary conditions, physics schemes, model modification), CAMx, UFS-CMAQ
- **Weather Forecast Models** (Advanced): WRF (NWP, nested domains, physics schemes), WRFDA (data assimilation), UFS Model
- **Emission Models** (Intermediate): VEIN, EmissV, SMOKE
- **Oceanic Models** (Intermediate): WRF-Chem-ROMS

Data Analysis & Processing

- **Emission Processing** (Advanced): WRF-Chem, CAMx, UFS-CMAQ
- **Observation Comparison** (Advanced): Model-observation comparison and statistical analysis
- **R and Python** (Advanced): Data processing, scripting, visualization, and analysis
- **Satellite and Specialized datasets** (Advanced): Processing, visualization and use of grided datasets (CERES, MODIS, MAIAC, SCIAMCHY, MOPITT, OMI, AIRS, MERGE, GPCP, PRISM, TMPA, TEMPO) and non-grided datasets (METAR, INMET, AQS, AERONET, Purple Air) to improve and validate models

Programmer & Git

- **R** (Expert): R package development, scripting and analysis
- **Python** (Intermediate): scripting and analysis
- **Fortran / C# / C++** (Intermediate): compiling and debugging
- **Bash / Csh** (Intermediate): Automation, model job submission
- **Git** (Intermediate): version control and collaborative development

Supercomputing

- **HPC** (Advanced): Globus, Derecho, Anvil, Discovery, Cheyenne, Stampede2
- **Job Schedulers & Workflow** (Intermediate): Slurm, PBS, Rococo

Languages

- **Portuguese** (Native)
- **English** (Professional)
- **Japanese / 日本語** (Beginner)
- **Spanish** (Beginner)

Projects

NOAA AC4: Integrating Atmospheric Chemistry and Carbon Cycle Research - Investigating atmospheric emissions and their impact on trace gases, aerosols, and Earth system processes

NYCI Project - Projecting New York's Air Quality to 2030 under Emission Controls.

EPA project - Multimodel Applications and Comparisons for Air Quality in the U.S..

iSuper - Intelligent Solutions to Urban Pollution for Equity and Resilience (iSUPER) pairs low-cost, adaptable sensing technologies with novel pollution prediction models to accurately identify hyperlocal pollution hot spots in real-time.

Wellcome Thrust UK - Development and application of WRF-Chem to simulations of the air pollution (ozone and particulate matter) at the street level for two major Brazilian metropolitan areas for the present day and future (2050).

PAPILA - Multi-model prediction of Air Pollution in Latin America and the Caribbean.

QUALAMET - Modeling the impact of vehicle emissions with different fuels on air quality using information from environmental measurements and emission inventories.

Pró-Estratégia 050/2011 - Development of an algorithm to simulate the dispersion of gases and particles released by space vehicles at the Alcântara Launch Center.

GILTT solution - Analytical solution and Simulation of pollutant dispersion with chemical reaction in the atmospheric planetary boundary layer using transform techniques.

Awards

2021 - XIV Prêmio AEA de Meio Ambiente (XIV Brazilian Association of Automotive Engineering Environmental Award) - Award for the QUALAMET project (Modeling of the Impact of Vehicle Emissions with Different Fuels on Air Quality) in the Environmental Responsibility category.

Competitive research funds

Comparative analysis of Air Quality scenarios for Brazil and downscaling for metropolitan areas for 2050 using WRF-Chem mode. 1,500,000 SUs on NCAR Derecho supercomputer system, award UNTE0008. 2024-08-22 to 2026-07-31.